

Volume I · Chapter 4 — Core Machine Learning for Medicine · Theory

Source: split from med-tech-curriculum-V1.md (V1). From this split onward this chapter is maintained **independently** here. See Volume-I-Chapter-4-context.md for the AI interaction log.

Prerequisites: Ch 1 (NumPy/Pandas), Ch 2 (clinical data).

Learning outcomes — the student can:

- Frame a clinical problem as **supervised or unsupervised** ML and build/train/evaluate models with **scikit-learn**.
- Apply a rigorous **workflow** (train/val/test, cross-validation) and avoid **data leakage**.
- Choose and **interpret clinically appropriate metrics** (sensitivity/specificity, PPV/NPV, ROC-AUC, PR-AUC, calibration).
- Use **unsupervised** methods (clustering, PCA) for **patient stratification**, and recognize imbalance/bias issues.

Topics (theory) — 14 h:

#	Topic	h
4.1	ML framing for medicine: supervised vs. unsupervised; clinical use cases	1
4.2	The ML workflow: train/validation/test, cross-validation, data leakage	2

#	Topic	h
4.3	Supervised I: logistic regression, linear models, regularization	2
4.4	Supervised II: decision trees, random forests, gradient boosting	2
4.5	Evaluation & metrics: confusion matrix, sensitivity/specificity/PV/NPV, ROC-AUC, PR-AUC, calibration	3
4.6	Class imbalance, thresholds & clinical decision trade-offs	1
4.7	Unsupervised: clustering (k-means, hierarchical) + PCA for stratification	2
4.8	Overfitting, bias/variance & model selection	1

Datasets/tools: synthetic structured EHR; scikit-learn, Pandas, Matplotlib.

Assessment: modeling project — build + evaluate a classifier with written interpretation (**50%**); evaluation/metrics lab rubric (**30%**); quiz (**20%**).

Key decisions: metric choice for the clinical question; **sensitivity vs. specificity** trade-off; complexity vs. **interpretability**; handling imbalance; avoiding leakage.

References: plan §13 → *Programming & data science*; *Clinical datasets*.

Hours: Theory **14** + Lab **16** = **30**. (*Feeds plan Lab 2 — Simple Clinical Classifier.*)